Exercises: Concentration Inequalities, LLN, CLT & Convergence – Answer Key

Mathematical Foundations for Machine Learning

Concentration Inequalities

Exercise 1.1 [Easy]

Let X be a random variable with $\mathbb{E}[X] = 5$ and $\mathrm{Var}(X) = 4$.

a) Using Chebyshev's inequality, find an upper bound for $P(|X - 5| \geq 3)$.

b) Using Chebyshev's inequality, find a value k such that $P(|X - 5| \geq k) \leq 0.1$.

Solution

a) Chebyshev's inequality states $P(|X - \mathbb{E}[X]| \geq k) \leq \frac{\mathrm{Var}(X)}{k^2}$. With $\mathrm{Var}(X) = 4$ and $k = 3$:

$$P(|X - 5| \geq 3) \leq \frac{4}{9} \approx 0.444.$$

b) We need $\frac{4}{k^2} \leq 0.1$, so $k^2 \geq 40$, giving $k \geq \sqrt{40} = 2\sqrt{10} \approx 6.32$.

Exercise 1.2 [Easy]

Let $X_1, \ldots, X_n$ be i.i.d. random variables with $X_i \in [0, 1]$ and $\mathbb{E}[X_i] = \mu$. Using Hoeffding's inequality, find the smallest n such that

$$P\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right| \geq 0.05\right) \leq 0.02.$$

Solution

Hoeffding's inequality gives:

$$P\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right| \geq t\right) \leq 2\exp\left(\frac{-2n^2t^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\right).$$

With $a_i = 0$, $b_i = 1$, $t = 0.05$:

$$2\exp(-2n \cdot 0.0025) \leq 0.02 \implies \exp(-0.005n) \leq 0.01 \implies n \geq \frac{\ln(100)}{0.005} = \frac{4.605}{0.005} = 921.$$

So we need $n \geq 921$.

Exercise 1.3 [Medium]

Let $X \sim \mathcal{N}(0, \sigma^2 I_d)$ be a d-dimensional Gaussian random vector and let $Z = \|X\|^2$.

a) Compute $\mathbb{E}[Z]$ and $\mathrm{Var}(Z)$.

b) Using Chebyshev's inequality, show that

$$P\left(\left|\|X\|^2 - d\sigma^2\right| > t\sqrt{d}\,\sigma^2\right) \leq \frac{2}{t^2}.$$

c) What does this tell us about the distribution of $\|X\|$ as $d \to \infty$? Express your answer in terms of where the mass of X concentrates.

Solution

a) Write $X = (X_1, \ldots, X_d)$ with $X_i \sim \mathcal{N}(0, \sigma^2)$ independent. Then $Z = \sum_{i=1}^d X_i^2$. Since $\mathbb{E}[X_i^2] = \sigma^2$ and $\mathrm{Var}(X_i^2) = \mathbb{E}[X_i^4] - (\mathbb{E}[X_i^2])^2 = 3\sigma^4 - \sigma^4 = 2\sigma^4$:

$$\mathbb{E}[Z] = d\sigma^2, \qquad \mathrm{Var}(Z) = 2d\sigma^4.$$

b) Applying Chebyshev with $k = t\sqrt{d}\,\sigma^2$:

$$P(|Z - d\sigma^2| > t\sqrt{d}\,\sigma^2) \leq \frac{\mathrm{Var}(Z)}{(t\sqrt{d}\,\sigma^2)^2} = \frac{2d\sigma^4}{t^2 d\sigma^4} = \frac{2}{t^2}.$$

c) The bound shows that $\|X\|^2$ concentrates around $d\sigma^2$ with relative fluctuations of order $1/\sqrt{d}$. As $d \to \infty$, the mass of X concentrates on a thin shell of radius $\approx \sigma\sqrt{d}$. Almost all probability mass lies in $\|X\| \in [\sigma\sqrt{d} - O(1), \sigma\sqrt{d} + O(1)]$.

Exercise 1.4 [Medium]

Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli random variables with $P(X_i = 1) = p$ and $P(X_i = 0) = 1 - p$, and let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$.

a) Derive upper bounds on $P(|\bar{X}_n - p| \geq \varepsilon)$ using both Chebyshev's and Hoeffding's inequality.

b) For $\varepsilon = 0.1$ and $p = 0.5$, determine numerically for which values of n the Hoeffding bound is tighter than the Chebyshev bound.

Solution

a) *Chebyshev:* We have $\mathrm{Var}(\bar{X}_n) = \frac{p(1-p)}{n}$. Chebyshev gives:

$$P(|\bar{X}_n - p| \geq \varepsilon) \leq \frac{p(1-p)}{n\varepsilon^2}.$$

Hoeffding: Since $X_i \in [0, 1]$:

$$P(|\bar{X}_n - p| \geq \varepsilon) \leq 2\exp(-2n\varepsilon^2).$$

b) With $\varepsilon = 0.1$ and $p = 0.5$: Chebyshev gives $\frac{0.25}{0.01n} = \frac{25}{n}$. Hoeffding gives $2e^{-0.02n}$. Setting them equal:

$$2e^{-0.02n} = \frac{25}{n}.$$

Numerically, the crossover is around $n \approx 54$. For $n \leq 53$, Chebyshev is tighter; for $n \geq 54$, Hoeffding is tighter. This illustrates that Hoeffding's exponential decay dominates for large n (high confidence regime), while Chebyshev can be better for small n.

Exercise 1.5 [Hard]

Let $X \sim \mathcal{N}(0, \sigma^2 I_d)$ and define $Z = \|X\|^2$.

a) Show that $\mathbb{E}[\langle X, AX\rangle] = \sigma^2 \mathrm{tr}(A)$ for any symmetric matrix $A \in \mathbb{R}^{d\times d}$.

b) Let $B \in \mathbb{R}^{m\times d}$. Using part (a), show that $\mathbb{E}[\|BX\|^2] = \sigma^2\|B\|_F^2$.

c) Using Chebyshev's inequality and part (b), derive a bound on

$$P\left(\left|\frac{1}{d}\|BX\|^2 - \frac{\sigma^2\|B\|_F^2}{d}\right| > \varepsilon\right)$$

in terms of d, σ, $\|B\|_F$, $\|B\|_{\text{op}}$ and ε. You may use $\text{Var}(\|BX\|^2) \leq 2\sigma^4\|B\|_F^2\|B\|_{\text{op}}^2$ without proof.

Solution

a) Write $\langle X, AX\rangle = \sum_{i,j} a_{ij}X_iX_j$. Taking expectation and using $\mathbb{E}[X_iX_j] = \sigma^2\delta_{ij}$:

$$\mathbb{E}[\langle X, AX\rangle] = \sum_{i,j} a_{ij}\sigma^2\delta_{ij} = \sigma^2\sum_i a_{ii} = \sigma^2\text{tr}(A).$$

b) We have $\|BX\|^2 = X^TB^TBX = \langle X, B^TBX\rangle$. Applying part (a) with $A = B^TB$:

$$\mathbb{E}[\|BX\|^2] = \sigma^2\text{tr}(B^TB) = \sigma^2\|B\|_F^2,$$

where the last step uses $\text{tr}(B^TB) = \sum_{i,j} b_{ij}^2 = \|B\|_F^2$.

c) Let $W = \frac{1}{d}\|BX\|^2$. Then $\mathbb{E}[W] = \frac{\sigma^2\|B\|_F^2}{d}$ and $\text{Var}(W) = \frac{1}{d^2}\text{Var}(\|BX\|^2) \leq \frac{2\sigma^4\|B\|_F^2\|B\|_{\text{op}}^2}{d^2}$. By Chebyshev:

$$P(|W - \mathbb{E}[W]| > \varepsilon) \leq \frac{2\sigma^4\|B\|_F^2\|B\|_{\text{op}}^2}{d^2\varepsilon^2}.$$

If B has bounded operator norm, the bound decays as $O(1/d)$ since $\|B\|_F^2 = O(d)$ for typical matrices, showing concentration in high dimensions.

Law of Large Numbers & Central Limit Theorem

Exercise 2.1 [Easy]

Let $X_1, X_2, \ldots$ be i.i.d. random variables with $\mathbb{E}[X_i] = 3$ and $\text{Var}(X_i) = 12$.

a) What does the Law of Large Numbers tell us about $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$ as $n \to \infty$?

b) Using the Central Limit Theorem, state the approximate distribution of $\bar{X}_n$ for large n.

c) Approximate $P(\bar{X}_{100} > 3.5)$ using the CLT. You may use $\Phi(1.44) \approx 0.925$, where Φ is the standard normal CDF.

Solution

a) The LLN states that $\bar{X}_n \to \mathbb{E}[X_i] = 3$ in probability as $n \to \infty$. That is, for any $\varepsilon > 0$, $P(|\bar{X}_n - 3| > \varepsilon) \to 0$.

b) By the CLT, $\sqrt{n}\frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{d} \mathcal{N}(0,1)$. So for large n:

$$\bar{X}_n \approx \mathcal{N}\left(3, \frac{12}{n}\right).$$

c) For $n = 100$: $\bar{X}_{100} \approx \mathcal{N}(3, 0.12)$. Standardizing:

$$P(\bar{X}_{100} > 3.5) = P\left(Z > \frac{3.5 - 3}{\sqrt{0.12}}\right) = P\left(Z > \frac{0.5}{0.3464}\right) \approx P(Z > 1.44).$$

Using $\Phi(1.44) \approx 0.925$: $P(\bar{X}_{100} > 3.5) \approx 1 - 0.925 = 0.075$.

Exercise 2.2 [Medium]

A biased die has probability distribution: $P(X = k) = \frac{k}{21}$ for $k = 1, 2, \ldots, 6$.

a) Compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

b) If the die is rolled $n = 200$ times, use the CLT to approximate the probability that the sample mean exceeds 4.5. You may use $\Phi(1.61) \approx 0.946$.

c) Using Chebyshev's inequality, find the minimum number of rolls n needed to guarantee that $P(|\bar{X}_n - \mathbb{E}[X]| \geq 0.5) \leq 0.05$.

Solution

a) $\mathbb{E}[X] = \sum_{k=1}^{6} k \cdot \frac{k}{21} = \frac{1}{21}\sum_{k=1}^{6} k^2 = \frac{91}{21} = \frac{13}{3} \approx 4.333$.
$\mathbb{E}[X^2] = \sum_{k=1}^{6} k^2 \cdot \frac{k}{21} = \frac{1}{21}\sum_{k=1}^{6} k^3 = \frac{441}{21} = 21$.
$\mathrm{Var}(X) = 21 - \left(\frac{13}{3}\right)^2 = 21 - \frac{169}{9} = \frac{189-169}{9} = \frac{20}{9} \approx 2.222$.
b) By CLT, $\bar{X}_{200} \approx \mathcal{N}\left(\frac{13}{3}, \frac{20/9}{200}\right) = \mathcal{N}\left(\frac{13}{3}, \frac{1}{90}\right)$.

$$P(\bar{X}_{200} > 4.5) = P\left(Z > \frac{4.5 - 13/3}{\sqrt{1/90}}\right) = P\left(Z > \frac{0.1\overline{6}}{0.1054}\right) \approx P(Z > 1.61) \approx 1 - 0.946 = 0.054.$$

c) Chebyshev: $P(|\bar{X}_n - \frac{13}{3}| \geq 0.5) \leq \frac{\mathrm{Var}(X)}{n \cdot 0.25} = \frac{20/9}{0.25n} = \frac{80}{9n}$. Setting $\frac{80}{9n} \leq 0.05$ gives $n \geq \frac{80}{0.45} \approx 177.8$, so $n \geq 178$.

Exercise 2.3 [Hard]

Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Define $S_n = \sum_{i=1}^{n} X_i$.

a) Show using Chebyshev's inequality that $\bar{X}_n = \frac{S_n}{n}$ converges to μ in probability (this is the Weak Law of Large Numbers).

b) Let $g : \mathbb{R} \to \mathbb{R}$ be a continuous function. Using the CLT, show that for large n:

$$\mathbb{E}[g(\bar{X}_n)] \approx g(\mu) + \frac{\sigma^2}{2n} g''(\mu).$$

Hint: Use a second-order Taylor expansion of g around μ and the CLT approximation.

c) Apply part (b) to estimate $\mathbb{E}[e^{\bar{X}_n}]$ when $\mu = 0$ and $\sigma^2 = 1$, and compare it to the exact value e^{μ} as $n \to \infty$.

Solution

a) By Chebyshev: $P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}$. As $n \to \infty$, the right-hand side $\to 0$, so $\bar{X}_n \xrightarrow{P} \mu$.
b) Taylor expand g around μ: $g(\bar{X}_n) \approx g(\mu) + g'(\mu)(\bar{X}_n - \mu) + \frac{1}{2}g''(\mu)(\bar{X}_n - \mu)^2$. Taking expectations:

$$\mathbb{E}[g(\bar{X}_n)] \approx g(\mu) + g'(\mu)\underbrace{\mathbb{E}[\bar{X}_n - \mu]}_{=0} + \frac{1}{2}g''(\mu)\underbrace{\mathbb{E}[(\bar{X}_n - \mu)^2]}_{=\mathrm{Var}(\bar{X}_n)=\sigma^2/n}.$$

This gives $\mathbb{E}[g(\bar{X}_n)] \approx g(\mu) + \frac{\sigma^2}{2n}g''(\mu)$.
c) With $g(x) = e^x$, $\mu = 0$, $\sigma^2 = 1$: $g(0) = 1$, $g''(0) = 1$. So $\mathbb{E}[e^{\bar{X}_n}] \approx 1 + \frac{1}{2n}$. As $n \to \infty$, $\mathbb{E}[e^{\bar{X}_n}] \to e^0 = 1$, consistent with LLN ($\bar{X}_n \to 0$). The correction $\frac{1}{2n}$ captures the bias from Jensen's inequality: since e^x is convex, $\mathbb{E}[e^{\bar{X}_n}] > e^{\mathbb{E}[\bar{X}_n]}$, and the gap shrinks as $O(1/n)$.

Convergence of Random Variables

Exercise 3.1 [Easy]

State whether each of the following implications is true or false. Give a brief justification.

a) Convergence in probability implies convergence in distribution.

b) Convergence in distribution implies convergence in probability.

c) If $X_n \xrightarrow{P} c$ for a constant c, does $X_n \xrightarrow{d} c$?

Solution

a) True. This is a standard result. If $P(|X_n - X| > \varepsilon) \to 0$ for all $\varepsilon > 0$, then $F_{X_n}(x) \to F_X(x)$ at all continuity points of F_X.

b) False in general. Counterexample: let $X \sim \mathcal{N}(0,1)$ and define $X_n = -X$ for all n. Then $X_n \xrightarrow{d} X$ (since $-X$ has the same distribution as X), but $P(|X_n - X| > \varepsilon) = P(|2X| > \varepsilon) \neq 0$, so convergence in probability fails.

c) True. When the limit is a constant, convergence in probability and convergence in distribution are equivalent. This is because $F_c(x) = \mathbf{1}_{x \geq c}$ has only one discontinuity at $x = c$, and convergence in probability to c directly implies $P(X_n \leq x) \to 0$ for $x < c$ and $P(X_n \leq x) \to 1$ for $x > c$.

Exercise 3.2 [Medium]

Let $U_1, U_2, \ldots$ be i.i.d. Uniform$(0,1)$ random variables and define $M_n = \max(U_1, \ldots, U_n)$.

a) Show that $P(M_n \leq x) = x^n$ for $x \in [0,1]$.

b) Determine whether M_n converges in probability to some limit. If so, identify the limit.

c) Find the distribution of $Y_n = n(1 - M_n)$ as $n \to \infty$. What type of convergence does this represent?

Solution

a) Since U_i are i.i.d.:

$$P(M_n \leq x) = P(U_1 \leq x, \ldots, U_n \leq x) = \prod_{i=1}^{n} P(U_i \leq x) = x^n$$

for $x \in [0,1]$.

b) For any $\varepsilon > 0$: $P(|M_n - 1| \geq \varepsilon) = P(M_n \leq 1 - \varepsilon) = (1-\varepsilon)^n \to 0$ as $n \to \infty$. So $M_n \xrightarrow{P} 1$.

c) For $y \geq 0$:

$$P(Y_n > y) = P(n(1 - M_n) > y) = P\Big(M_n < 1 - \frac{y}{n}\Big) = \Big(1 - \frac{y}{n}\Big)^n \to e^{-y}$$

as $n \to \infty$. Therefore $P(Y_n \leq y) \to 1 - e^{-y}$, which is the CDF of an Exp(1) distribution. This is **convergence in distribution**: $Y_n \xrightarrow{d} \text{Exp}(1)$.

Exercise 3.3 [Hard]

Let $X_1, X_2, \ldots$ be i.i.d. Uniform$(0,1)$ and define

$$Y_n = \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \frac{1}{3}\right).$$

a) Compute $\mathbb{E}[X_i^2]$ and $\mathrm{Var}(X_i^2)$.
b) Using the CLT, determine the asymptotic distribution of Y_n.
c) Use part (b) to approximate $P(Y_{400} > 0.05)$. You may use $\Phi(0.47) \approx 0.681$.
d) Is it possible to apply Hoeffding's inequality to bound $P\left(\left|\frac{1}{n}\sum_{i=1}^n X_i^2 - \frac{1}{3}\right| \geq t\right)$? If so, state the bound.

Solution

a) $\mathbb{E}[X_i^2] = \int_0^1 x^2 dx = \frac{1}{3}$.
$\mathbb{E}[X_i^4] = \int_0^1 x^4 dx = \frac{1}{5}$.
$\mathrm{Var}(X_i^2) = \frac{1}{5} - \frac{1}{9} = \frac{4}{45}$.
b) Define $Z_i = X_i^2$. Then Z_i are i.i.d. with $\mathbb{E}[Z_i] = \frac{1}{3}$ and $\mathrm{Var}(Z_i) = \frac{4}{45}$. By the CLT:

$$Y_n = \sqrt{n}\left(\bar{Z}_n - \frac{1}{3}\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{4}{45}\right).$$

c) $Y_{400} \approx \mathcal{N}(0, 4/45)$, so $\mathrm{sd}(Y_{400}) = \frac{2}{\sqrt{45}} \approx 0.2981$. Then

$$P(Y_{400} > 0.05) \approx P\left(Z > \frac{0.05}{2/\sqrt{45}}\right) = P\left(Z > \frac{0.05\sqrt{45}}{2}\right) = P(Z > 0.168).$$

Since $\sqrt{45} = 3\sqrt{5} \approx 6.708$, we get $\frac{0.05 \cdot 6.708}{2} \approx 0.168$. Using standard tables, $\Phi(0.17) \approx 0.567$, so $P(Y_{400} > 0.05) \approx 1 - 0.567 = 0.433$.
d) Yes. Since $X_i \in [0,1]$, we have $Z_i = X_i^2 \in [0,1]$, so Z_i are bounded i.i.d. random variables. Hoeffding's inequality applies:

$$P\left(\left|\bar{Z}_n - \frac{1}{3}\right| \geq t\right) \leq 2\exp(-2nt^2).$$

Confidence Intervals for the Test Error

Exercise 4.1 [Easy]

A classifier f is evaluated on $n = 1000$ i.i.d. test samples and achieves an empirical test error of $\hat{\mathrm{err}}_f = 0.08$.
a) Using Hoeffding's inequality, construct a 95% confidence interval for the true error err_f.
b) How many test samples are needed to guarantee the confidence interval has width at most 0.02 at 95% confidence?

Solution

a) From Hoeffding, with probability $\geq 1 - \delta$:

$$|\hat{\mathrm{err}}_f - \mathrm{err}_f| \leq \sqrt{\frac{1}{2n}\log\frac{2}{\delta}}.$$

With $\delta = 0.05$ and $n = 1000$:

$$\sqrt{\frac{\log(40)}{2000}} = \sqrt{\frac{3.689}{2000}} \approx \sqrt{0.001844} \approx 0.0429.$$

So the 95% CI is $[0.08 - 0.043,\ 0.08 + 0.043] = [0.037,\ 0.123]$.

b) We need the half-width ≤ 0.01 (so total width ≤ 0.02):

$$\sqrt{\frac{\log(40)}{2n}} \leq 0.01 \implies n \geq \frac{\log(40)}{2 \cdot 0.0001} = \frac{3.689}{0.0002} = 18445.$$

So we need $n \geq 18{,}445$ test samples.

Exercise 4.2 [Medium]

Let f be a binary classifier with true error $\mathrm{err}_f = P(f(X) \neq Y)$, evaluated on n i.i.d. test samples.

a) Write $\hat{\mathrm{err}}_f$ in terms of the indicator variables $\mathbf{1}_{f(X_i)\neq Y_i}$ and show that $\mathrm{Var}(\hat{\mathrm{err}}_f) = \frac{\mathrm{err}_f(1-\mathrm{err}_f)}{n}$.

b) Using the fact that $\mathrm{err}_f(1-\mathrm{err}_f) \leq \frac{1}{4}$, derive a confidence interval via Chebyshev's inequality.

c) For $n = 10{,}000$ and $\varepsilon = 0.01$, compare the upper bounds from Chebyshev and Hoeffding on $P(|\hat{\mathrm{err}}_f - \mathrm{err}_f| \geq \varepsilon)$. Which one is tighter?

Solution

a) $\hat{\mathrm{err}}_f = \frac{1}{n}\sum_{i=1}^n \mathbf{1}_{f(X_i)\neq Y_i}$. Let $Z_i = \mathbf{1}_{f(X_i)\neq Y_i}$. Then $Z_i \sim \mathrm{Bernoulli}(\mathrm{err}_f)$ are i.i.d., so:

$$\mathrm{Var}(\hat{\mathrm{err}}_f) = \frac{1}{n^2}\sum_{i=1}^n \mathrm{Var}(Z_i) = \frac{n \cdot \mathrm{err}_f(1-\mathrm{err}_f)}{n^2} = \frac{\mathrm{err}_f(1-\mathrm{err}_f)}{n}.$$

b) By Chebyshev:

$$P(|\hat{\mathrm{err}}_f - \mathrm{err}_f| \geq \varepsilon) \leq \frac{\mathrm{err}_f(1-\mathrm{err}_f)}{n\varepsilon^2} \leq \frac{1}{4n\varepsilon^2}.$$

So with confidence $\geq 1-\delta$: $|\hat{\mathrm{err}}_f - \mathrm{err}_f| \leq \frac{1}{2\sqrt{n\delta}}$.

c) With $n = 10{,}000$ and $\varepsilon = 0.01$:

Chebyshev: $\frac{1}{4 \cdot 10{,}000 \cdot 0.0001} = \frac{1}{4} = 0.25$.

Hoeffding: $2e^{-2 \cdot 10{,}000 \cdot 0.0001} = 2e^{-2} \approx 2 \cdot 0.1353 = 0.271$.

In this case Chebyshev (0.25) is slightly tighter than Hoeffding (0.271). This is a regime where $n\varepsilon^2$ is moderate. For larger $n\varepsilon^2$, Hoeffding's exponential decay will dominate.

Exercise 4.3 [Hard]

You are evaluating two classifiers f and g on the same test set of n i.i.d. samples. Define $\Delta = \mathrm{err}_f - \mathrm{err}_g$ and $\hat{\Delta} = \hat{\mathrm{err}}_f - \hat{\mathrm{err}}_g$.

a) Let $Z_i = \mathbf{1}_{f(X_i)\neq Y_i} - \mathbf{1}_{g(X_i)\neq Y_i}$. Express $\hat{\Delta}$ in terms of Z_i and note that $Z_i \in [-1, 1]$.

b) Using Hoeffding's inequality on Z_i, derive a confidence interval for Δ.

c) If $n = 5000$, $\hat{\mathrm{err}}_f = 0.12$ and $\hat{\mathrm{err}}_g = 0.10$, determine with 95% confidence whether there is a statistically significant difference between the two classifiers.

Solution

a) $\hat{\Delta} = \hat{\mathrm{err}}_f - \hat{\mathrm{err}}_g = \frac{1}{n}\sum_{i=1}^n \left(\mathbf{1}_{f(X_i)\neq Y_i} - \mathbf{1}_{g(X_i)\neq Y_i}\right) = \frac{1}{n}\sum_{i=1}^n Z_i$.
Each Z_i takes values in $\{-1, 0, 1\}$, and $\mathbb{E}[Z_i] = \mathrm{err}_f - \mathrm{err}_g = \Delta$.

b) Since $Z_i \in [-1, 1]$ are i.i.d., Hoeffding gives:

$$P(|\hat{\Delta} - \Delta| \geq \varepsilon) \leq 2\exp\left(\frac{-2n^2\varepsilon^2}{n \cdot 4}\right) = 2\exp\left(\frac{-n\varepsilon^2}{2}\right).$$

With probability $\geq 1 - \delta$:

$$|\hat{\Delta} - \Delta| \leq \sqrt{\frac{2}{n}\log\frac{2}{\delta}}.$$

c) With $\delta = 0.05$ and $n = 5000$:

$$\sqrt{\frac{2\log(40)}{5000}} = \sqrt{\frac{7.378}{5000}} \approx \sqrt{0.001476} \approx 0.0384.$$

The observed $\hat{\Delta} = 0.02$, so the 95% CI for Δ is $[0.02 - 0.038,\ 0.02 + 0.038] = [-0.018,\ 0.058]$. Since $0 \in [-0.018, 0.058]$, we **cannot** conclude a statistically significant difference at the 95% confidence level.

Mixed Challenging Exercises

Exercise 5.1 [Hard]

A machine learning engineer trains a model and wants to evaluate it rigorously. The test set has n i.i.d. samples, each in $\mathbb{R}^d$, with labels in $\{0, 1\}$.

a) The model achieves $\hat{\text{err}}_f = 0.03$ on $n = 500$ test samples. Using Hoeffding's inequality, compute a 99% confidence interval for err_f.

b) The engineer claims: "My model has at most 5% error." Using part (a), can this claim be supported at 99% confidence?

c) How large must n be so that a 99% confidence interval using Hoeffding has half-width at most 0.01?

d) Using the CLT, approximate the distribution of $\hat{\text{err}}_f$ for $n = 500$ assuming $\text{err}_f = 0.03$. Compare the CLT-based 99% confidence interval to the one from Hoeffding. You may use $z_{0.005} = 2.576$.

Solution

a) With $\delta = 0.01$ and $n = 500$:

$$\sqrt{\frac{\log(2/0.01)}{2 \cdot 500}} = \sqrt{\frac{\log(200)}{1000}} = \sqrt{\frac{5.298}{1000}} \approx 0.0728.$$

99% CI: $[0.03 - 0.073,\ 0.03 + 0.073] = [0,\ 0.103]$ (clamped to $[0, 0.103]$ since error ≥ 0).

b) The upper bound of the CI is $0.103 > 0.05$. So the claim cannot be supported at 99% confidence — the interval includes values above 5%.

c) Need $\sqrt{\frac{\log(200)}{2n}} \leq 0.01$, so $n \geq \frac{5.298}{2 \cdot 0.0001} = 26{,}490$.

d) By CLT, $\hat{\text{err}}_f \approx \mathcal{N}\left(0.03, \frac{0.03 \cdot 0.97}{500}\right) = \mathcal{N}(0.03, 5.82 \times 10^{-5})$. The standard deviation is $\sqrt{5.82 \times 10^{-5}} \approx 0.00763$.

CLT-based 99% CI: $0.03 \pm 2.576 \cdot 0.00763 = 0.03 \pm 0.0197$, i.e., $[0.010,\ 0.050]$.

The CLT interval $[0.010, 0.050]$ is *much* tighter than the Hoeffding interval $[0, 0.103]$, because (i) the CLT uses the actual variance $\text{err}_f(1 - \text{err}_f)$ rather than the worst-case $1/4$, and (ii) the Gaussian approximation exploits the shape of the distribution rather than just its

boundedness. However, the CLT interval requires knowledge of err_f (or plugging in $\hat{\mathrm{err}}_f$), while Hoeffding's bound is distribution-free.

Exercise 5.2 [Hard]

Consider a dataset of n i.i.d. samples $X_1, \ldots, X_n$ drawn uniformly from $[0,1]^d$.

a) For a fixed point $p \in [0,1]^d$, compute $\mathbb{E}[\|X_i - p\|^2]$ and show that it does not depend on p when $p = (\frac{1}{2}, \ldots, \frac{1}{2})$. *Hint:* Use $\mathrm{Var}(X_{i,j}) = \frac{1}{12}$ for $X_{i,j} \sim \mathrm{Uniform}(0,1)$.

b) Using Hoeffding's inequality applied coordinate-wise, show that

$$P\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_{i,j} - \frac{1}{2}\right| \geq t\right) \leq 2e^{-2nt^2}$$

for each coordinate j.

c) Using a union bound and part (b), find a value t such that with probability $\geq 1-\delta$, the sample mean $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ satisfies $\|\bar{X}_n - (\frac{1}{2}, \ldots, \frac{1}{2})\|_\infty \leq t$.

d) Interpret the result: how does the number of samples n needed to achieve $\|\bar{X}_n - \mu\|_\infty \leq \varepsilon$ with confidence $1-\delta$ scale with the dimension d?

Solution

a) With $p = (\frac{1}{2}, \ldots, \frac{1}{2})$: $\mathbb{E}[\|X_i - p\|^2] = \sum_{j=1}^{d} \mathbb{E}[(X_{i,j} - \frac{1}{2})^2] = \sum_{j=1}^{d} \mathrm{Var}(X_{i,j}) = \frac{d}{12}$, since $\mathbb{E}[X_{i,j}] = \frac{1}{2}$ and $\mathrm{Var}(X_{i,j}) = \frac{1}{12}$.

b) For each coordinate j, $X_{1,j}, \ldots, X_{n,j}$ are i.i.d. with $X_{i,j} \in [0,1]$ and $\mathbb{E}[X_{i,j}] = \frac{1}{2}$. Direct application of Hoeffding's inequality (with $a_i = 0$, $b_i = 1$) gives:

$$P\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_{i,j} - \frac{1}{2}\right| \geq t\right) \leq 2e^{-2nt^2}.$$

c) By the union bound over d coordinates:

$$P\big(\|\bar{X}_n - \mu\|_\infty \geq t\big) \leq \sum_{j=1}^{d} P\left(|\bar{X}_{n,j} - \frac{1}{2}| \geq t\right) \leq 2d \cdot e^{-2nt^2}.$$

Setting $2d \cdot e^{-2nt^2} \leq \delta$:

$$t = \sqrt{\frac{1}{2n}\log\frac{2d}{\delta}}.$$

d) To achieve $t \leq \varepsilon$, we need $n \geq \frac{1}{2\varepsilon^2}\log\frac{2d}{\delta}$. The sample complexity scales as $O(\log(d)/\varepsilon^2)$. The logarithmic dependence on d is mild — even in very high dimensions, the sample mean converges uniformly across coordinates with only a logarithmic price in d. This is a key insight: concentration in the ℓ_∞ norm degrades gracefully with dimension.